ClaimSense AI · Checkpoint 2 Final Submission

FDE Spine Capstone · Final Deliverable (Milestones 1·5)

Participant Name: Amirdha Gopal
Tech Stack: AWS + OpenAI Agents SDK / Ollama + Python 3.12 (FastAPI) + Docker + Terra
form + Vanilla Web UI
Submission Type: Checkpoint - 2 (Final Submission: Milestones 1 through 5)
Date: 23 October 2026
Final Release Tag: `v1.0.0` (Git commit `bcffc18`)
Repository URL: https://github.com/amirdhagopal/claimsense

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1. Executive Summary & Required Assets

This document certifies the successful completion of all five milestones of the Claim
Sense AI Motor Claims Adjudication Co-Pilot capstone project for Apex Motor Insurance Ltd.

Three Required Submission Assets:

Asset | Location / Value | Description
Asset 1: Final Codebase | https://github.com/amirdhagopal/claimsense
Tag: `v1...
Asset 2: Observation Document | [`docs/observation-document.md`](https://github.co...
Asset 3: Deployed Frontend + Demo | Frontend: `http://localhost:8000/dashboard` <b...

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2. Asset 1: Final Codebase Verification

· Repository: `https://github.com/amirdhagopal/claimsense`
· Release Tag: `v1.0.0`
· Local Test Suite: 39 passed in 18.31s (`PYTHONPATH=. pytest tests/ -v`)
· Evaluation Suite: 150/150 cases passed schema validation and automated benchmark ch
ecks (`python3 evals/run_evals.py`)

Git History Milestone Trace:

1. `3b4e914`: Milestone 1 · Discovery, Charter, RACI, Success Criteria, 20 Baseline C
ases.
2. `f23555f`: Milestone 2 · Statement of Work, Pricing & Unit Economics, Typed Contra
cts, Terraform IaC.
3. `1431b73`: Milestone 3 · 4-Agent Pipeline, BM25 Retrieval, Deterministic Spine, Do
cker, CI Gate.
4. `7c09fd7`: Milestone 4 · Route-Level Bearer Auth, Two-User Access Control Isolatio
n Test, Audit Trail.
5. `bcffc18`: Milestone 5 · 150-Case Evals, Operational Dashboard, Runbook, Incident
Report, Tag `v1.0.0`.

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3. Asset 2: Completed Observation Document Summary

The complete observation document is checked into [`docs/observation-document.md`](ht
tps://github.com/amirdhagopal/claimsense/blob/main/docs/observation-document.md). Key findings across its 6 required sections include:

1. Production Deployment & Readiness Assessment:

Validated containerized FastAPI service on ECS Fargate / local Docker.
Deterministic code boundary completely separates access control and date/money ma
th from foundation models.
2. Evaluation Suite Results & Performance Analysis:
150 ground-truth cases across all 4 outcomes (49 Approve, 40 Deny, 32 Flag, 29 Ro
ute).
33 adversarial cases across 4 attack vectors (prompt injection, cross-record leak
age, forced auto-decision, narrative contradictions) successfully blocked with 0% adversarial bypass.
3. Observability, Telemetry & SLA Compliance:
CloudWatch-compatible metric pipeline tracking total call volume, error rate, and
p50/p95/p99 latencies.
SLA Alarms configured: `HighLatencyP95Alarm` (> 60s) and `HighErrorRateAlarm` (>
5%).
4. Platform Reliability & Failure Modes (SEV-2 Incident Drill):
Simulated an upstream model latency breach (> 60s p95) under load.
Alarm fired in 30 seconds; ECS auto-rollback initiated; system recovered to healt
hy state in 2 minutes 45 seconds MTTR. Full report in [`docs/incident-report.md`](https://github.com/amirdhagopal/claimsense/blob/main/doc
5. Operational Handover & Production Readiness:
4 detailed Standard Operating Procedures in [`docs/runbook.md`](https://github.co
m/amirdhagopal/claimsense/blob/main/docs/runbook.md):
- SOP-01: Provisioning a New Claims Adjuster User.
- SOP-02: Rolling Back a Production Deployment on AWS ECS.
- SOP-03: Escalating a Fraud-Flagged Claim to SIU.
- SOP-04: Updating the Policy Clause Corpus and Re-indexing BM25.
6. Retrospective & Key Engineering Lessons Learned:
Strict Pydantic contracts and deterministic code boundaries are essential in regu
lated domains.
Smaller models require explicit output sanitization (e.g. handling markdown fence
s and empty exclusion arrays).

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4. Asset 3: Deployed Frontend & Demo Recording Guide

4.1 Deployed Web Co-Pilot URL

· URL: `http://localhost:8000/dashboard` (also accessible at root `http://localhost:8
000/`)
· Technology: FastAPI backend serving an interactive, glassmorphic dark-mode web appl
ication (HTML5/CSS3/Vanilla JS).

4.2 Interactive Co-Pilot Capabilities

1. Preset Claim Dossier Switcher: Instant loading of standard collision claims (`CLM-
2026-0101`), adversarial prompt injection attempts (`CLM-ADV-001`), cross-record leakage attempts (`CLM-ADV-002`), or total loss claims.
2. One-Click Real-Time Adjudication: Executes the 4-agent pipeline (`FileReader`, `Co
verage`, `RiskAssessment`, `DecisionDrafter`) through the deterministic spine and displays token breakdown, BM25 retrieved clauses, and f
3. Adjuster Override & Decision Write-Back: Human adjusters can edit authorized payou
ts, enter adjuster rationale, and commit an idempotent decision record back to the system.
4. Live Operational Telemetry: Real-time polling of call volume, CPM, error rate, p50
/p95 latencies, and CloudWatch SLA alarm status.
5. Interactive Audit Log & Evals Viewer: In-browser inspection of tamper-proof audit
trails and the 150-case evaluation suite distribution.

4.3 Demo Walkthrough Sequence (for video/live demo)

1. Step 1 (0:00 - 0:45): Open `http://localhost:8000/dashboard`. Point out the system
   health badge and live CloudWatch telemetry metrics.
2. Step 2 (0:45 - 1:30): Select claim `CLM-2026-0101`. Click "Adjudicate Claim with C
   laimSense AI". Show the deterministic spine passing in 1ms, followed by the 4 agents extracting facts, matching clause `CLS-PER-006` via
3. Step 3 (1:30 - 2:15): In the Adjuster Write-Back panel, modify the authorized payo
   ut from $2,300 to $2,100, add the note "Adjusted labor rate based on regional tariff", and click "Confirm Decision". Show the confirmed badg
4. Step 4 (2:15 - 3:00): Select adversarial claim `CLM-ADV-001` (Prompt Injection). R
   un adjudication. Demonstrate how the system resists the injection and properly flags the claim.
5. Step 5 (3:00 - 3:30): Navigate to the Observability and Evaluation tabs to show li
   ve metrics and the 150-case benchmark.

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5. Form Submission Fields Quick Reference

Use these values when completing the Google Form:

- Name: `Amirdha Gopal`
- Cohort Name: (Enter your cohort name, e.g. FDE Spine Capstone)
- Tech Stack Used: `AWS + OpenAI Agents SDK / Ollama + Python 3.12 (FastAPI) + Docker
  + Terraform + Vanilla Web UI`
- Select the Checkpoint submission: `Checkpoint - 2`
- Submission Link (Upload File): Upload `docs/ClaimSense_AI_Checkpoint_2_Final_Submis
  sion.pdf` (compiled from this document).